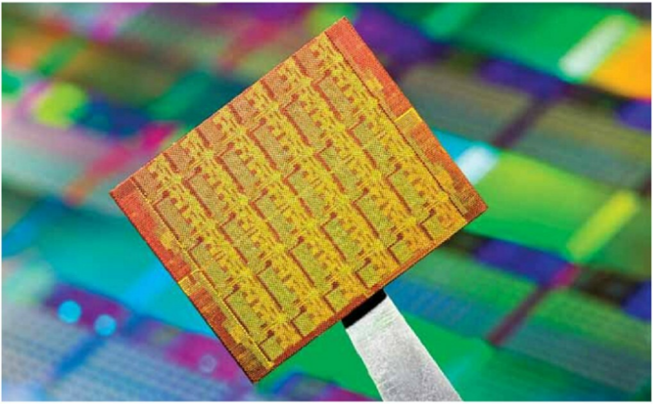

Intel's single-chip cloud computer (SCC) has 48 Intel cores and runs at as low as 25 watts

SCC. However, this chip seems more complex, with a much larger die size, system-level complexities and the challenges of a 45nm physical design to boot. The success of this chip will make processors more scalable than you ever thought possible, and also accelerate multiple-core software research and advanced development.

According to the company, future laptops with processing capability of this magnitude could have 'vision' similar to humans who can see objects and motion as it happens, with high accuracy. Imagine, for example, a laptop with a 3-D camera and display giving you virtual dance lessons or showing you a 'mirror' of yourself wearing the clothes you want to buy online. Twirl and turn to watch how the fabric drapes and how its colour complements your skin tone. This kind of interaction could eliminate the need for keyboards, remote controls or joysticks in gaming. Some researchers believe computers may even be able to read brain waves, so simply thinking about a command, such as the words you want typed, would execute it without the need for speech.

We can expect more breakthroughs from Intel's stable, considering the amount of research being done in this space. "On the hardware side, we explore scalable multi-core architecture that integrates streamlined processor cores and accelerators using a fast, energy-efficient, modular core-to-core infrastructure. Our research is also focused on memory sharing and stacking to provide a high-bandwidth, flexible cache and memory hierarchy that supports many simultaneous threads, fairly and efficiently; and on high-bandwidth I/O and communications that balance compute demands with the I/O and network demands, within the platform power and cost budgets," lists Erranguntla.

The soft side of success

As far as multi-core computing goes, more than half the reason for success depends on the ability of the program to efficiently harness the immense parallel computing power of the processor. "To take advantage of the increasing number of cores, efficient load balancing of software is required. In addition, we need to identify and come up with programs and applications for these systems, to improve performance. It is also important for OEMs to strike a balance between performance and power consumption. We at AMD understand this and are working with our partners to address these issues," says Vamsi Krishna.

Intel, too, has taken several initiatives to educate developers on the specific challenges and techniques of parallel computing, in order to enable them to make good use of multi-core systems.

The multi-core related software research vision at Intel includes "...model-based applications that use tera-scale capabilities to comprehend data, make smarter decisions, and make visual experiences look, act and feel real; parallel programming tools that empower the ordinary programmer to develop applications that use parallelism with scalability and performance, safety and reliability; and thread-aware execution environments that provide real-time performance and power management across cores, and scale with increasing thread and